Group Reflection Journal

ITAI 1371 | Lab 06: Regression and Classification | NeuralCrew

Patrick Ray

In Lab 6, I learned the difference between regression and classification models. Regression is used to predict continuous numerical values, while classification is used to predict categories or classes. In this lab, I used Linear Regression to predict passenger fares from the Titanic dataset using Age and Pclass. I also used Logistic Regression to predict whether a passenger survived using Age, Pclass, and Sex.

For the Linear Regression model, I trained the model using the training data, made predictions using the test data, and evaluated the model using mean squared error. My model produced a Mean Squared Error of 3364.92. I also examined the model coefficients and learned that they show how each feature affects the predicted fare. The Age coefficient was approximately -0.44, and the Pclass coefficient was approximately -38.30.

For the Logistic Regression model, I encoded the Sex column into numerical values, trained the model, made predictions, and evaluated the results using accuracy. The model achieved an accuracy of 74.83% when predicting passenger survival.

One challenge I experienced during the lab was receiving a NaN error while training the Logistic Regression model. I learned that the Sex column had been encoded more than once after rerunning the cell. I corrected the problem by reloading the original dataset and running the notebook cells in the proper order. This helped me understand why the order in which code cells are executed is important when working in a Jupyter notebook.

Overall, this lab helped me better understand how regression and classification solve different machine learning problems. It also gave me more practice with preparing data, training models, making predictions, evaluating model performance, and troubleshooting errors.

Individual Contribution – Patrick Ray

Viviana Zamora

Reflection summary based solely on Viviana’s submitted contribution statement, used at her request; no separate personal reflection was supplied.

Viviana reported completing and testing both Linear Regression and Logistic Regression, correcting notebook formatting, training and evaluating both models, reviewing their outputs and coefficients, and answering the knowledge-check questions on regression, classification, coefficients, mean squared error, and accuracy.

Funsho Orogun

During this lab, I learned how to build, train, and evaluate basic machine learning models using scikit-learn for both regression and classification tasks. Working through the exercises helped me better understand the differences between these two types of predictive models and when each should be used.

One of the most important things I learned is that regression and classification solve different kinds of problems. Regression is used to predict continuous numerical values, such as Titanic passenger fares, while classification is used to predict categories, such as whether a passenger survived or not.

I also learned that these models require different evaluation methods. Regression models are commonly measured using Mean Squared Error (MSE), which shows how far predictions are from the actual values, while classification models are evaluated based on accuracy, which measures how often the predicted class is correct. This helped me understand why accuracy is not a good metric for regression problems, since even a prediction that is very close to the actual value would still be considered incorrect.

Another concept I learned was how to interpret the coefficients of a Linear Regression model. By examining the .coef_ attribute, I saw how the model assigns a weight to each feature, indicating how much the predicted outcome changes when that feature changes while the other features remain constant. This provided valuable insight into how the model makes its predictions.

I also gained an important lesson in troubleshooting and debugging Python code. During the lab, I encountered several SyntaxError issues caused by import statements being accidentally combined on the same line. Through this experience, I learned that Python requires clear separation between commands and that each import statement should be placed on its own line. Once I corrected the formatting, the code executed successfully. This reinforced the importance of clean, organized code and careful attention to syntax when working on data science and machine learning projects.

Overall, this lab strengthened both my understanding of machine learning concepts and my ability to identify and fix coding errors, making me more confident in developing and evaluating predictive models.

Dustin Mastrota (Levi)

In this lab, I used Google Colab to train and evaluate linear regression and logistic regression models using the Titanic dataset. The linear regression model predicted a passenger's fare based on their age and passenger class, while the logistic regression model predicted survival using age, passenger class, and sex. Working with both models in the same notebook helped me connect the difference between predicting a numerical value and predicting a category. One of my main takeaways was that evaluating a model depends on what it is predicting. For fare prediction, I used mean squared error to measure the differences between the predicted and actual fares. For survival prediction, I used accuracy to measure how often the model predicted the correct outcome. The regression model produced an MSE of 3364.92, and the classification model achieved an accuracy of 74.83%. Seeing these different results helped me understand why the same evaluation method would not be appropriate for both tasks. Another part of the lab that stood out was preparing the data before training the models. The notebook removed rows with missing ages and converted the sex values into numerical values. I also learned how the linear regression coefficients describe the relationship between each feature and the predicted fare while the other feature stays constant. This helped me think about the model as more than just code that produces a prediction. Looking back at the output, I noticed that the square root of the MSE was 58.01, but the printed explanation still said "about $48." That detail reminded me to check the explanations alongside the calculated results instead of assuming that a successful code run means everything is correct. Overall, this lab helped me connect data preparation, model training, prediction, and evaluation into one workflow.

Compiled from members' submitted materials. Individual statements are attributed; Viviana's section is a source-based summary rather than an independently submitted reflection.